

NGUYEN HUY HOANG

AI Engineer

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OBJECTIVES

Short-term (1–3 years):

- Develop and integrate LLMs (Claude, OpenAI, Gemini) into production systems, focusing on RAG pipelines, semantic search, and real-world Q&A applications.
- Master evaluation frameworks and optimization techniques to ensure AI output quality, latency, and cost-efficiency across document-based retrieval systems.
- Build end-to-end AI features: smart search, content summarization, information extraction with measurable quality metrics and continuous monitoring.

Long-term (3–5 years):

- Become a senior AI Engineer specializing in scalable Generative AI, multimodal systems, and AI-driven automation.
- Drive measurable business value through MLOps best practices, model reliability, and deep domain understanding.

TECHNICAL SKILLS

- **Machine Learning:** Supervised/Unsupervised Learning, Math & Statistics Foundations, Feature Engineering
- **Deep Learning:** Computer Vision, NLP, LLM, Transfer Learning, Fine-tuning
- **Generative AI & LLM:** Prompt Engineering, Embeddings, RAG, AI Agents, MCP (Model Context Protocol), Stable Diffusion
- **MLOps & Tooling:** CI/CD, Docker, FastAPI, REST API, Git, Experiment Tracking, n8n, Model Monitoring
- **Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, Scikit-learn
- **Data & Vector Stack:** NumPy, Pandas, FAISS, Vector Databases, Data Preprocessing Pipelines
- **Cloud & Platforms:** Azure OpenAI, Azure AI Search, Microsoft Copilot Studio, Power Automate, SharePoint, Google Colab
- **Programming Languages:** Python, C++, SQL

WORK EXPERIENCES

Fresher AI Engineer – TDI Group

02/2026 – 05/2026

Ha Noi, Vietnam

Customer Search & Support Chatbot (Copilot Studio + MCP + RAG)

- Deployed an intelligent customer support chatbot with RAG architecture, achieving 90%+ accuracy on 100+ real-world queries across 5 product categories through systematic performance evaluation.
- Integrated Model Context Protocol (MCP) connecting 3+ internal data sources; optimized latency from 9.6s to 3.4s via parallel data fetching and response caching.
- Automated document/image retrieval (50+ assets in SharePoint via Power Automate), reducing manual ticket handling by 35% and enabling 24/7 self-service support.
- Built vector search pipeline with dense embeddings and FAISS; reduced hallucination rate through 3-layer prompt engineering (context grounding, reasoning separation, disclaimers), validated on 100+ edge cases.
- Developed enterprise Q&A system supporting 6+ document formats and video sources; tested on 300+ Vietnamese pairs; delivered responses with source citations and auto-generated follow-ups.

Handwritten Text Recognition & Spell Correction System

- Improve an end-to-end Vietnamese HTR pipeline (PyTorch CRNN+CTC, ~9.3M params) on 15,354 training lines; reached 8.78% CER / 25.82% WER on held-out test and 6.40% / 20.04% on validation, with a ~2.4 pp CER val-test gap indicating controlled overfitting.
- Added paragraph line segmentation (horizontal projection profile) and end-to-end inference; 17.21% CER / 43.35% WER on 30 paragraph images (~30 ms/line on Apple Silicon MPS).
- Explored spell correction via n-gram language models and edit-distance post-processing to reduce WER on noisy OCR output.
- Reviewed CRNN-CTC, attention, and transformer OCR; compared against VietOCR baseline to guide model and experiment design.
- Built unified write/ + UIT_HWDB_line indexing and augmentation pipeline (affine, perspective, noise, morphology); documented results and reproduction steps for institute research reporting.

PROJECTS

Enterprise RAG Knowledge Platform

02/2026 – 04/2026

Research Project · Python, FastAPI, LangChain, Claude Sonnet 4, Voyage AI, Qdrant

<https://github.com/huyhung9630/trungtamtrithuc>

- Built a full-stack enterprise Q&A system supporting multi-format ingestion (PDF, DOCX, XLSX, TXT, MD, MP4, YouTube, Playlist), returning Vietnamese answers with source citations and AI-generated follow-up suggestions.
- Designed a 3-tier PDF parsing pipeline (Docling → Claude Vision → pdfplumber) with automated table detection and LLM-based table description; implemented heading-aware chunking (700 tokens, overlap 80) and Voyage AI 1024-dim vector embeddings stored on Qdrant Cloud.
- Engineered a hybrid 3-layer conversation memory (sliding window + rolling summary + cross-session vector recall) with 4-layer anti-bloat guards - filtering ~60–80% of low-value turns without additional LLM calls.
- Applied Claude 4 prompt engineering best practices (XML-tagged context, quote-first grounding, reasoning lane) to minimize hallucination on complex document queries; added AI Auto Metadata generation via Claude Haiku + Pydantic tool use to eliminate manual tagging at upload time.
- Containerized with Docker, exposed RESTful API via FastAPI, and implemented GDPR-compliant memory deletion endpoints.

Dental Marketing Multi-Agent System

04/2025 – 06/2025

Research Project · Python, LangGraph, FastAPI, OpenAI / Claude / Gemini, Facebook Graph API

<https://github.com/datng1/Multi-Agents-SmileUp>

- Built a multi-agent marketing automation system with a CMO agent orchestrating a 10-step pipeline (Crawler → Insight → Trend → Strategy → Copywriting → Compliance → Publish), with API key priority fallback (Gemini → OpenAI → Anthropic) and local heuristic fallback to keep workflow running without credentials.
- Designed a CMO Jury scoring system (100-point scorecard, 7 dimensions) with 3-model consensus vote; content only proceeds on score ≥ 85 + compliance gate cleared - blocking absolute medical claims and missing consultation disclaimers.
- Integrated Meta Ad Library scraping (80/20 competitor-to-keyword ratio); generated two content tracks (ads_effective for conversion, page_care for trust-building) with unique run_seed per run to prevent hook/caption reuse.
- Built multi-page Facebook publishing with per-user page permissions, approval token signing, in-browser watermarking, and DRY_RUN mode; deployed via GitHub Actions CI/CD with 7-day per-user run history.

Research Paper Clustering System for Academic Conferences

09/2024 – 12/2024

Faculty Scientific Research (2nd Prize, 2024–2025) · Python, Scikit-learn, NLP

- Implemented an unsupervised K-Means clustering pipeline with TF-IDF embeddings to categorize conference papers, improving topic identification, session structuring, and reviewer allocation.
- Awarded 2nd Prize at faculty-level scientific research competition (2024–2025).

EDUCATION

Artificial Intelligence and Data Science – Thuyloi University

2022 – 2026

- GPA: 3.49 / 4.0
- Scholarship: Merit-based Scholarship for Academic Excellence
- Young Innovator Challenge – Hanoi University of Science and Technology: Top 10
- Second Prize – Faculty-level Scientific Research (2024–2025): "Clustering Researchers Using K-Means"